COMP1870 Theoretical Foundations of Computer Science I

School of Computer Science, University of Leeds

Worksheet - Week 1.4

This worksheet contains a combination of *formative activities* (which contribute towards your learning) and *summative activities* (which you will complete and submit to be assessed as part of your portfolio).

- Exercises marked with red borders are summative and must be submitted in Gradescope.
- Other exercises may be covered in tutorial sessions.
- Exercises marked by (*) are advanced, and may take some time to complete.
- Exercises marked by (*E*) are exam-style questions. You should expect to see questions like this in the exam.

Expectations:

1. **Timeliness** You should complete all of the activities in the order provided and submit your portfolio evidence on PebblePad before the deadline.
2. **Presentation:** You should present all of your work clearly and concisely following any additional guidance provided by the module staff.
3. **Integrity:** You are responsible for ensuring that the evidence you submit as part of your portfolio is **entirely your own work**. You can find out more about Academic integrity on the Skill@library website. All work you submit for assessment is subject to the academic integrity policy.

Feedback: Feedback on formative activities will be provided via Lab classes and tutorials. Feedback on evidence submitted as part of the portfolio will be available on Gradescope.

Support opportunities: Support with the activity sheet is available in tutorials and the Math Lunch Hour. Individual support is available via office hours.

Expected completion date: Friday 30 October 2025

All portfolio tasks are rated AMBER by the university: AI tools can be used in an assistive role. You are permitted to use AI tools for specific defined processes within the assessment. Within this assessment you may use Generative AI to:

- help you start to understand more complex ideas by providing accessible summaries.
- test your knowledge against a piece of content.
- help identify and correct spelling mistakes and grammatical errors in your work.
- provide feedback and advice on your overall coding style.

You must not use Gen AI to:

- produce written content (code, comments or any other content) which you then submit as your work, regardless of whether you make changes to it.
- rewrite or make changes to any of your work.

Learning outcomes

On completion of this activity sheet you will have:

1. demonstrated an understanding of the basic vocabulary of functions,
2. properly defined a function, identifying the domain, codomain, and range,
3. identified properties of functions and made arguments about them,
4. performed operations on functions, such as composition and inverse,
5. demonstrated an understanding of the basic vocabulary of relations, including reflexive, symmetric, transitive, and antisymmetric relations, total orders, partial orders, and equivalence relations,
6. identified properties of relations and made arguments about them.

Tutorials on Wednesday and Thursday will cover material from Week 4, Lecture 1. This material is in exercises 1–6. Tutorials on Monday and Tuesday will cover material from Week 4, Lecture 2. This material is in exercises 8–10.

Questions

1. Let A be the set $\{a, b, c, d, e, f\}$. For each of the following subsets of $A \times \mathbb{Z}$ decide whether it is a function from A to $\mathbb{Z}$. Explain your reasoning.
   (a) $\{(b, -4), (e, 0), (d, 1), (a, -1), (f, 0), (c, 9)\}$
   (b) $\{(c, 11), (f, 18), (d, -19), (a, -8), (f, 7), (e, 8)\}$
   (c) $\{(f, 0)\}$
   (d) $\{\}$
   (e) $\{(a, 5), (b, 5), (c, 6), (d, -1), (e, 9), (f, 101)\}$
2. (*E*) Let A be the set $\{1, 2, 3, 4\}$, let B be the set $\{u, v, w, x\}$, and let C be $\{\alpha, \beta, \gamma\}$. Let $f\colon A \to B$ and $g\colon B \to C$ be the following functions
$$f = \{(1, v), (2, x), (3, u), (4, x)\} \qquad g = \{(u, \alpha), (v, \beta), (w, \alpha), (x, \gamma)\}$$
   (a) Draw the arrow diagrams for f and g.
   (b) Write down the ordered pairs in the composition $g \circ f$.
3. (*E*) Let S be the set $\{1, 2, 3, 4\}$. Let $f\colon S \to S$ be the function
$$f\colon 1 \mapsto 3 \qquad f\colon 2 \mapsto 1 \qquad f\colon 3 \mapsto 4 \qquad f\colon 4 \mapsto 2$$
   (a) Draw arrow diagrams for f, f^{-1} (the inverse of f), and id_S (the identity function on S).
   (b) Use appropriate arrow diagrams to calculate the compositions $f \circ f^{-1}$, $f^{-1} \circ f$, $f \circ \mathrm{id}_S$, and $\mathrm{id}_S \circ f$.

4. (*E*) Let A be the set $\{\heartsuit, \diamondsuit, \clubsuit, \spadesuit\}$ and let B be the set $\{1, 2, 3, 4\}$. Consider the following subsets of $A \times B$. Which of them are functions? Of those that are functions, which are bijections?

   (a) $\{(\heartsuit, 1), (\clubsuit, 1), (\clubsuit, 1), (\spadesuit, 1)\}$
   (b) $\{(\heartsuit, 4), (\diamondsuit, 1), (\spadesuit, 2)\}$
   (c) $\{(\heartsuit, 3), (\diamondsuit, 1), (\clubsuit, 4), (\spadesuit, 2)\}$
   (d) $\{(\heartsuit, 1), (\diamondsuit, 4), (\clubsuit, 3), (\spadesuit, 3)\}$

5. (*E*) Consider the following functions. Decide whether each of them is injective, surjective, or bijective. Explain your reasoning.

   (a) $f\colon \mathbb{Z} \to \mathbb{Z}$ where $f(m) = 2m - 1$.
   (b) $f\colon \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ where $f(m, n)$ is the maximum of m and n.
   (c) $f\colon \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z} \times \mathbb{Z}$ where $f(m, n) = (n, m)$.

6. Consider the following dictionary data structure, which shows winners of the FIFA World Cup.

```
FIFA={"1994":"Brazil", "1998":"France", "2002":"Brazil", "2006":"Italy", "2010":"Spain",
      "2014":"Germany", "2018":"France", "2022":"Argentina", "2026":"Spain"}
```

   Draw an arrow diagram which shows this dictionary as a function.

7. Let A be the set $\{1, 2, 3, 4, 5, 6\}$ and let f and g be the following functions from A to A.

$$f = \{(1, 4), (2, 2), (3, 6), (4, 5), (5, 1), (6, 3)\} \qquad g = \{(1, 3), (2, 1), (3, 5), (4, 2), (5, 4), (6, 5)\}$$

   (a) Write down the ordered pairs in the inverse function f^{-1}.
   (b) Use your answer to (a) and function composition to find a function $h\colon A \to A$ such that $f \circ h = g$.

8. Let A be the set $\{a, b, c, d, e\}$. Consider the following relations on A.

$$R = \{(a, b), (b, c), (a, c), (a, d), (a, e), (b, d), (b, e), (c, d), (c, e), (d, e)\}$$
$$S = \{(a, a), (a, b), (b, a), (b, b), (c, c), (c, d), (d, c), (d, d), (d, e), (e, d), (c, e), (e, c), (e, e)\}$$
$$T = \{\}$$
$$U = \{(a, a), (b, b), (c, c), (d, d), (e, e)\}$$
$$V = \{(a, a), (a, b), (a, d), (a, e), (b, d), (b, e), (c, c), (c, d), (d, e)\}$$

   (a) Draw a directed graph diagram for each of R, S, T, U, and V.
   (b) For each of R, S, T, U, and V decide whether the relation is
      (i) reflexive,
      (ii) symmetric,
      (iii) anti-symmetric, and
      (iv) transitive.
   (c) Which of R, S, T, U, and V is an equivalence relation? For those that are equivalence relations, what are the equivalence classes?

(d) Which of R, S, T, U, and V is a partial order? For those that are partial orders, draw the corresponding Hasse diagram.

9. (*E*) Let A be the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$. Write down all the ordered pairs in the equivalence relation $R \subseteq A \times A$ such that the partition into equivalence classes is

$$\{\{1, 4, 7\}, \{2\}, \{3, 6\}, \{5, 8\}\}.$$

Draw the directed graph that represents R. Write down all the ordered pairs in R.

10. Let W be the set of all websites. Consider the relation $R \subseteq W \times W$ where $w_1 R w_2$ means that site w_1 contains a link to site w_2. Do you expect this relation to be reflexive, symmetric, anti-symmetric, or transitive?

11. Let A be the set $\{1, 2, 3, 4, 5\}$. Let $R \subseteq A \times A$ be the following relation on A.

$$\{(1, 1), (1, 3), (2, 2), (2, 4), (3, 3), (3, 5), (4, 4), (5, 5)\}.$$

(a) Draw the arrow diagram of R.

(b) Is R reflexive? Justify your answer.

(c) Is R symmetric? Justify your answer.

(d) Is R anti-symmetric? Justify your answer.

(e) Is R transitive? Justify your answer.

12. (*E*) Let A be the following set of sets.

$$A = \{\emptyset, \{1\}, \{2\}, \{3\}, \{2, 3, 4\}, \{1, 2, 5\}, \{1, 2, 3, 5\}\}.$$

Consider the relation $R \subseteq A \times A$ where XRY holds if and only if X is a subset of Y.

(a) Draw the directed graph of R.

(b) Draw the Hasse diagram of R.

13. (*) Let $f : A \to B$ and $g : B \to C$ be one-to-one functions. Prove that $g \circ f$ is one-to-one.

14. (*) Let $f : A \to B$ and $g : B \to C$ be onto functions. Prove that $g \circ f$ is onto.

15. (*) Let A be a set and let $R \subseteq A \times A$ be a relation on A. Prove that if R is reflexive, symmetric, and anti-symmetric, then $R = \{(a, a) \mid a \in A\}$.

16. (*) Let A be a set and let R be a relation on A. Define R^{-1} to be $\{(y, x) \mid (x, y) \in R\}$. Prove that if R is transitive then R^{-1} is also transitive.

17. (*) Let A be a set and let $\mathcal{P}$ be a partition of A. Define the relation R so that (a, b) is in R if and only if a and b belong to the same set in $\mathcal{P}$. Prove R is an equivalence relation.